

Design BMS.

- Overview
- Requirements
- Class Diagram
- Schema Design
- Code.

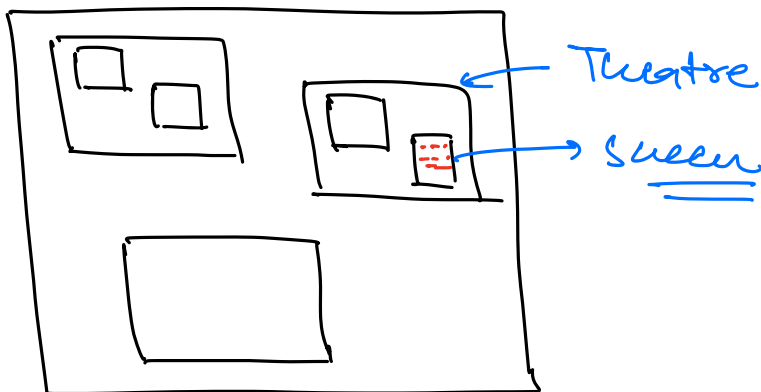
Overview.

- Alignment
- What to design?
 - Entity
 - Working Code.
- How to take input params?
- Do we need to persist the data?

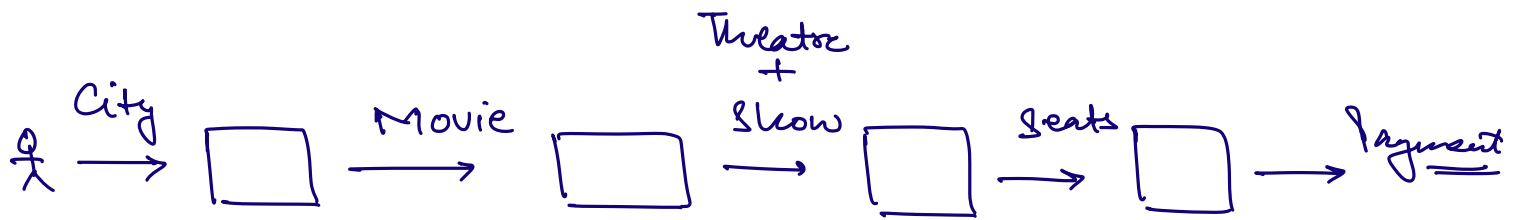
Requirements.

- 1) Users should be able book movie tickets.
- 2) Users can select seats while booking.
- 3) BMS has multiple supported regions | Cities.

City



- 4) Movie is being played inside a theatre
- 5) A theatre can have multiple screens, and one screen can run one movie at a time.
- 6) User books the ticket for a particular show.
- 7) A show is a movie running in a particular city at a particular theatre in a particular screen at fixed time
- 8) Users should be able to see all the movies running in that city.
- 9) Users should be able to see all the shows of a particular movie.

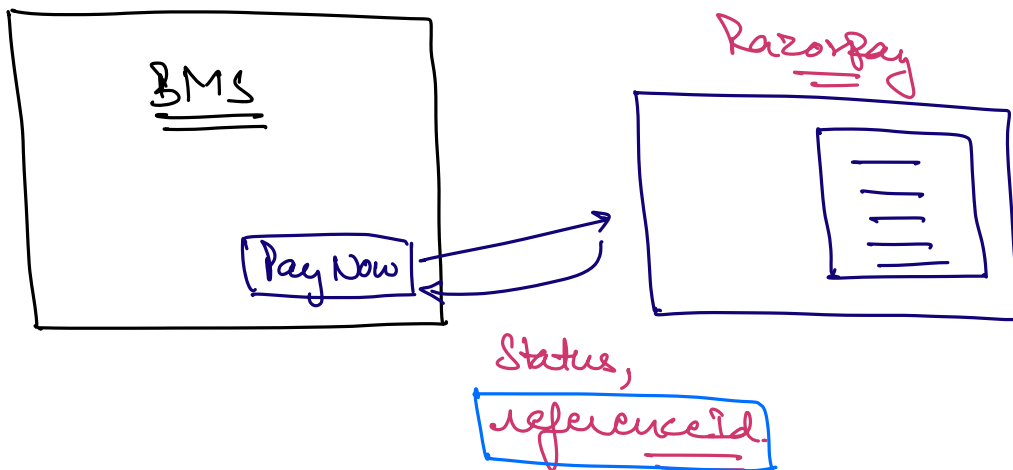


- 10) In one ticket, we can book upto 10 seats.
- 11) No 2 people should be able to book the same seat for the same show.

Price: Movie + Theatre + Screen + Type + language
+ time + Seat type

12) Price is an attribute of Show + Seat Type

13) Only online payment is supported, will be using 3rd party payment gateway for online payments.



14) Payment can be done via multiple mode of transactions.
(credit card, debit card, - - - -)

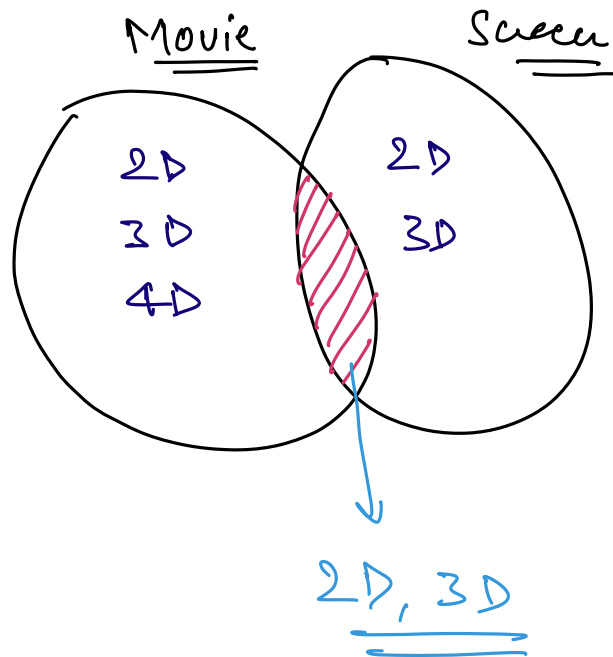
15) Partial payment is supported.

1000 ₹

- 200 ₹ wallet / Coupon
- 800 ₹ cc / - -

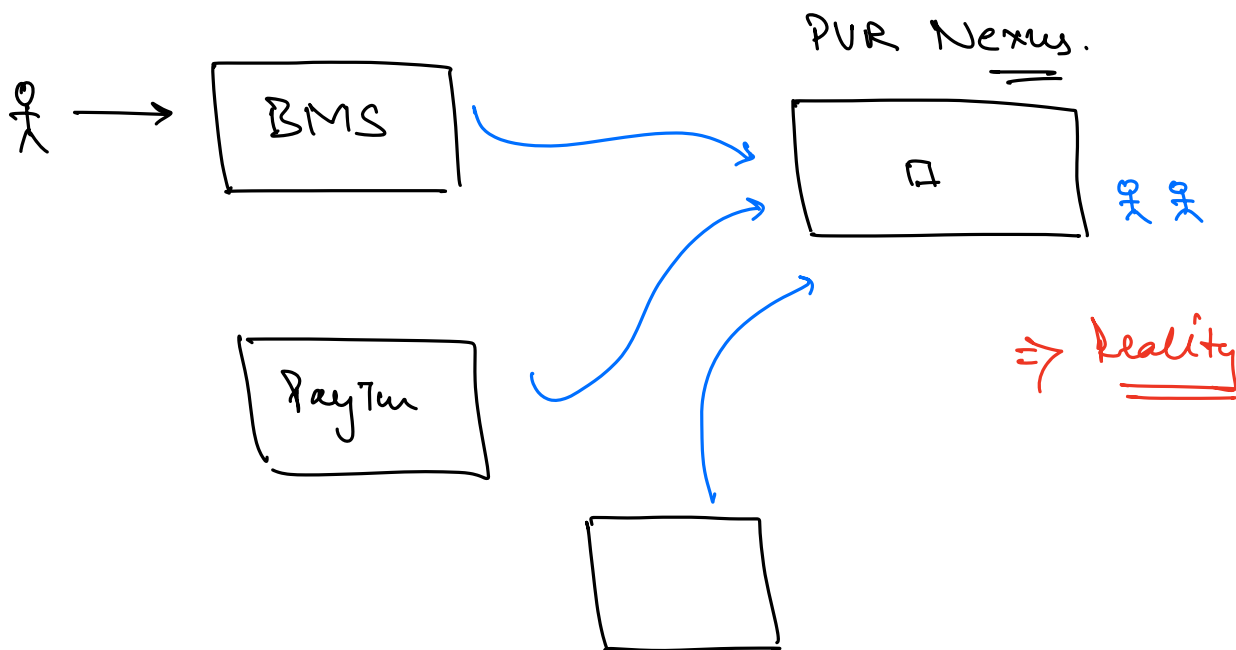
16) Users can cancel the booking.

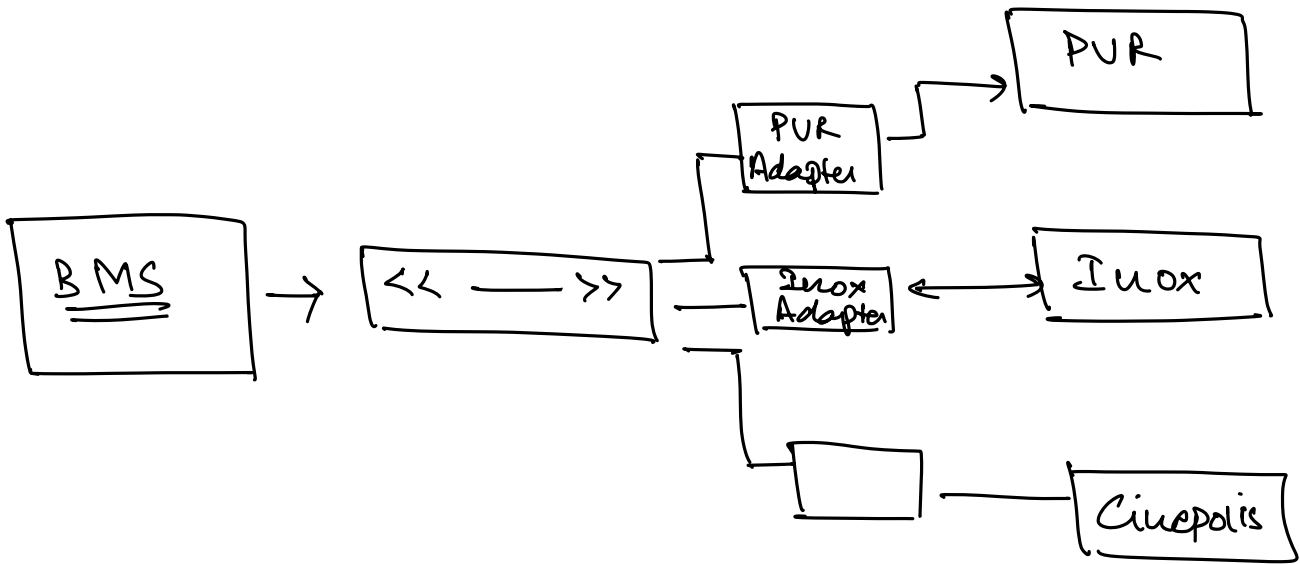
17) For every movie, we store its Cast, type, language, Genre...



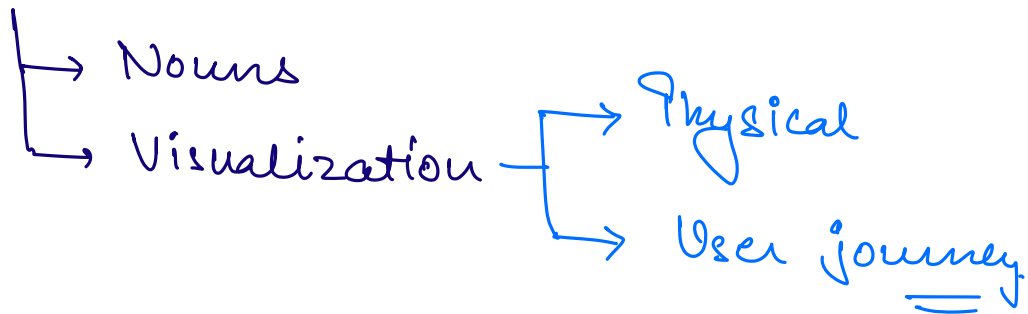
18) A movie can support multiple features
2D, 3D, 4D, IMAX, DOLBY,

19) A screen can also support multiple features.

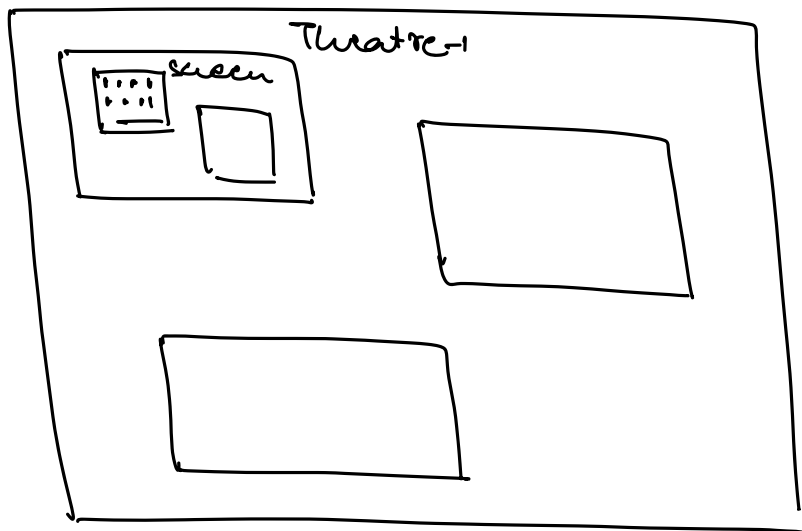




CLASS DIAGRAM.



Region/City



City
id
name
list<Theatre>

Theatre
id
name
list<Screen>
City

Screen
id
name
list<Seat>
list<features>

Feature
2D,
3D,
4D,
IMAX,
=

Seat
id
number
SeatType
Status
row
col

Status isn't an attribute of a Seat only. It is an attr of Show + Seat.

Status $\Rightarrow$ Show + Seat

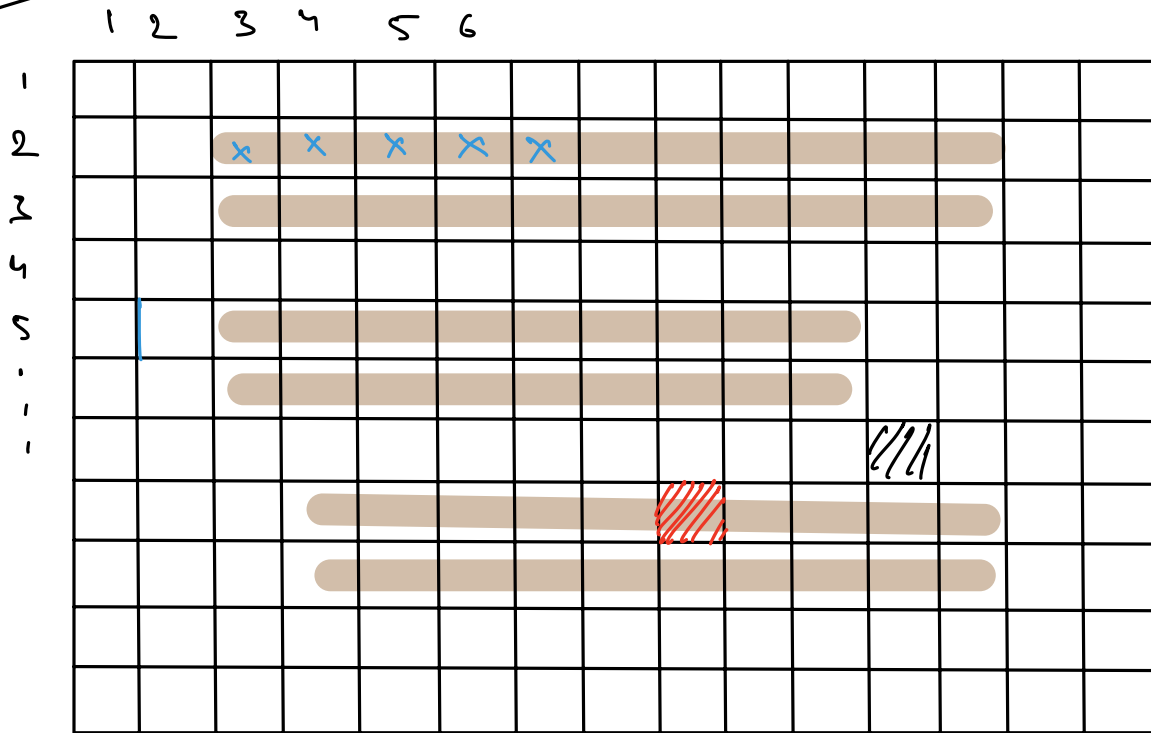
SeatType
GOLD,
SILVER,
PLATINUM



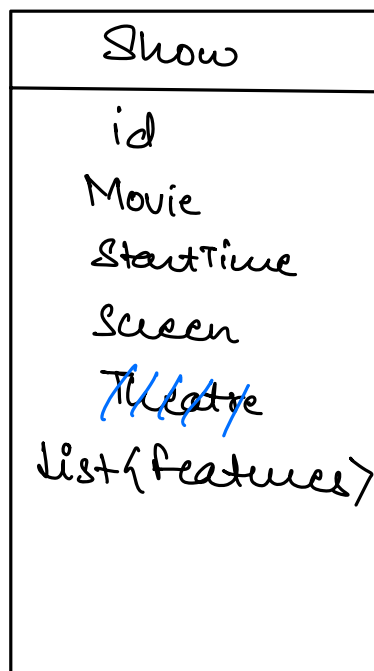
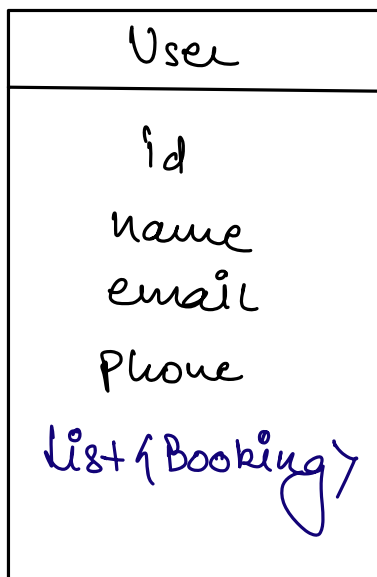
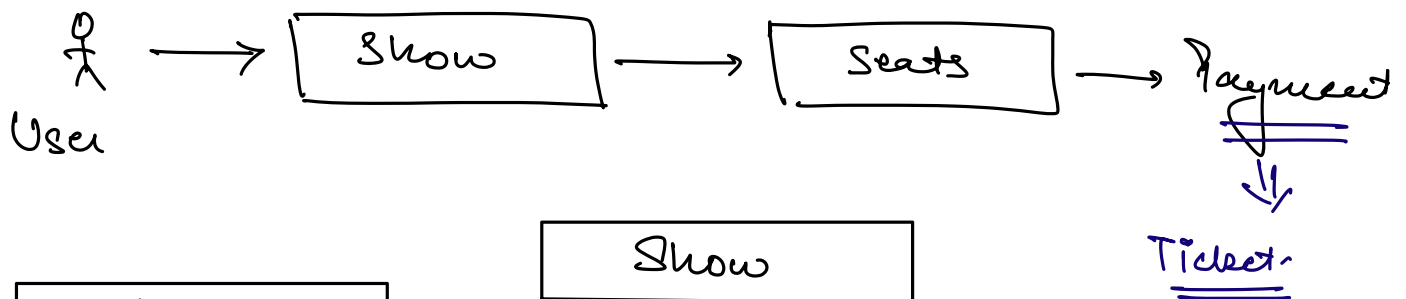
Class.

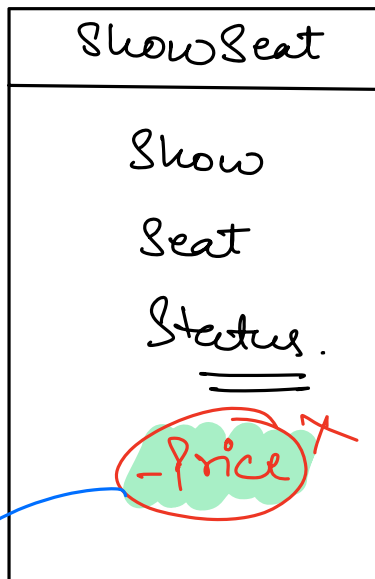
SeatType
id
<u>Value</u>

100x100



A1 $\begin{cases} 9:00\text{PM} \Rightarrow \text{Booked} \\ 6\text{PM} \Rightarrow \text{Available} \end{cases}$





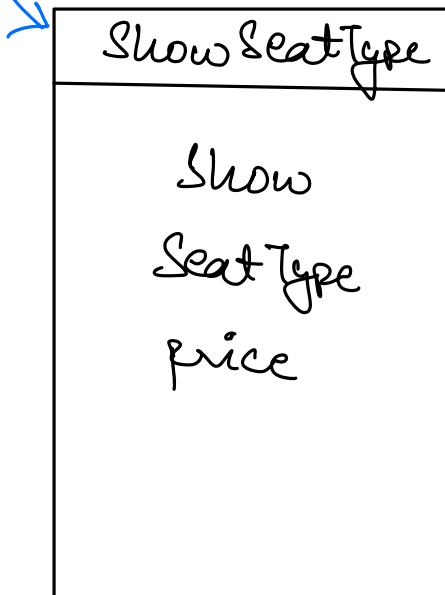
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Table: Show-Seat

Showid	SeatId	Status	Price
1	1	-	300
1	2	-	300
1	3	-	300
1	4	-	300
1	5	-	200
1	6	-	200
⋮	⋮	⋮	⋮

For a Show, all the seats of a particular type will have same Price.

Price: Show + SeatType



⇒

Table: Show-Seat-Type

Showid	Seat-type-id	Price
1	1	500
1	2	400
1	3	700

Booking Ticket
id
Status
User
list { ShowSeat }
time
amount
list { Payment }

Partial payments

Payment
id
amount
Status
referenceId
<u>PaymentMode</u>

PENDING
SUCCESS
FAILURE

UPI
CC
DC

— * —

⇒ Spring Boot.

⇒ Spring Data JPA